

ISHITA P ACHARYA

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EDUCATION

Mangalore Institute of Technology and Engineering (MITE) Nov 2022 – May 2026

Bachelor of Engineering in Computer Science - (Current GPA 7.92 / 10.0) Moodabidre, Karnataka, India

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, DBMS, Cloud Computing, Parallel Computing, Artificial Intelligence, Machine Learning, Software Engineering, Cryptography and Network Security.

SKILLS AND CERTIFICATIONS

Programming Languages: C, Python, JavaScript.

AI/ML & NLP: Scikit-learn, PyTorch, TensorFlow, Model Evaluation (Precision, Recall, F1-score), OpenCV, Tesseract OCR, Speech Recognition & Voice Processing, Text-to-Speech (TTS), Conversational AI, Image Preprocessing, MediaPipe, NumPy, Pandas, RAG, Vector Embeddings, Semantic Search, Hybrid Search, Prompt Engineering, Local LLM Deployment, Retrieval Pipelines, Isolation Forest, Anomaly Detection, Multi-Agent Systems

LLM & AI Tools: LangChain, Hugging Face, Hugging Face Inference Endpoints, Ollama, ChromaDB, Sentence Transformers, BM25 (rank-bm25), OpenAI API, ElevenLabs API, Spotify API, pdfplumber, PyMuPDF, Streamlit

Frameworks & Libraries: React.js, HTML5, CSS3, Bootstrap, Laravel, Node.js, Flask, FastAPI, REST APIs

Databases: MySQL, MongoDB, MongoDB Atlas.

Security: MITRE ATT&CK Framework, Log Analysis, Intrusion Detection, Threat Triage.

Cloud, DevOps & Systems: AWS EC2, S3, Ubuntu, Linux, Git, GitHub, Railway, REST API Design, JWT Authentication.

Certifications: Cyber Job Simulation (Delloite), Prompt Wars (MITE), Cybersecurity in the Product Development Lifecycle: From Design to Deployment (MITE), Cybersecurity: Vulnerability Assessment and Exploitation (MITE).

WORK EXPERIENCE

Software Engineer Intern | Udupi Web Solutions

Customer Support System CRM/ERP

Jan 2026 – May 2026

- * Developed and secured RESTful APIs using JWT-based authentication and role-based access control for Admin, Project Manager, Employee, and Client modules, ensuring efficient user management and secure data handling across the platform.
- * Built and optimized automated invoice generation workflows by integrating approved employee timesheets, reducing manual processing effort and improving billing accuracy and operational efficiency.
- * Designed and implemented scalable MongoDB database schemas to support multiple billing models, including Fixed Cost and Retainer-based systems, enabling flexible project and financial management.
- * Improved backend system performance by achieving sub-500ms API response times and maintaining over 70% test coverage using Jest, ensuring reliable, efficient, and well-tested application functionality.

ACADEMIC PROJECTS

Multi-PDF AI System

April 2026 - Present

Personal Project

- * Developed a privacy-focused Retrieval-Augmented Generation (RAG) application using Python, LangChain, Ollama, and ChromaDB, enabling intelligent question-answering across multiple PDF documents with source citations.
- * Implemented a hybrid retrieval pipeline combining BM25 keyword search, ChromaDB vector search, and Reciprocal Rank Fusion (RRF), improving document retrieval accuracy and contextual response quality.
- * Built an interactive Streamlit web application and CLI interface supporting PDF ingestion, document summarization, table extraction, and structured information retrieval from unstructured documents.
- * Integrated local LLM inference, embedding generation, and vector database indexing to provide secure, fully offline document intelligence without reliance on external cloud APIs.

AI Security Log & Incident Agent

June 2025 - June 2026

Personal Project

- * Built a multi-layer security log analysis pipeline in Python combining regex-based parsing (SSH/web/firewall logs), rule-based detection, and Isolation Forest anomaly detection, achieving 68% structuring coverage on mixed-format log data.
- * Integrated a LangChain + Hugging Face LLM agent to classify detected incidents against the MITRE ATT&CK taxonomy, assign severity, and generate plain-English analyst reports, with graceful degradation to rule-based fallback on LLM failure.
- * Diagnosed and resolved a deprecated third-party API integration mid-project by migrating to Hugging Face's current Inference Providers router, and hardened exception handling after testing exposed a gap in the system's fault-tolerance design.

Peripheral Arterial Disease Detection Using Machine Learning

March 2025 - November 2025

Team Project (4)

- * Built an end-to-end ML pipeline to detect Peripheral Arterial Disease (PAD) from unstructured Doppler reports by integrating OCR (Tesseract) with feature extraction and classification models.
- * Developed a Random Forest-based classification system with SMOTE to address class imbalance, with model accuracy of 75% and enhancing recall for minority classes.
- * Built a scalable Flask-based backend pipeline integrating OCR, ML inference, and rule-based validation for automated clinical report processing.
- * Deployed a Flask-based web application for real-time report analysis, enabling automated extraction, prediction, and clinically interpretable results.

ZOE AI Voice Assistant

February 2025

Personal Project

- * Developed an AI-powered voice assistant inspired by JARVIS using Python and Flask.
- * Integrated Speech Recognition to convert voice commands into text for real-time interaction.
- * Used OpenAI API to generate intelligent conversational responses and answer user queries.
- * Implemented ElevenLabs API for realistic text-to-speech output, creating a natural voice interaction experience.